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Too Cold to Count

A Matched Case-Control Study of Park Barbecue Units and the Usage Counter That Ranks Them for Renewal

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The problem

Public barbecue units are ranked for renewal by recorded usage, and the counter increments only on a completed heat cycle. A unit that no longer heats therefore records no demand. Is the ranking reading demand, or condition?

Research questions

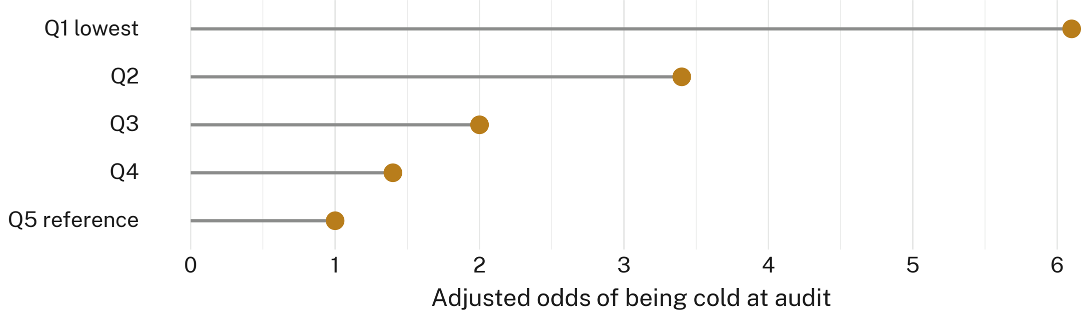
- Are cold units concentrated in the lowest recorded-usage quintiles?
- Does the association survive adjustment for reserve visitation?
- What share of cold units reaches the renewal shortlist?

Study design

- 248 units · 96 reserves · unannounced probe audit · 14 weeks
- 82 cases (cold at audit) matched 1:2 to working units on visitation band and installation cohort
- Conditional logistic regression; the cold threshold (below 60 °C at 90 seconds) was pre-registered and not adjusted post hoc

What we found

One third of the network was cold at audit. Recorded usage appears to track condition rather than demand: the least-used quintile returns an adjusted odds of being cold of 6.1 against the highest, and the gap widens with time since the unit's last completed cycle.



Adjusted odds of a unit being found cold at unannounced audit, by recorded-usage quintile over the preceding 12 months (82 matched sets across 96 reserves, audited March–June 2026). The lowest-use quintile returns 6.1 (95% CI 2.9–12.8) against the highest.

Interpretation

The counter sits downstream of the fault it is used to rank. Renewal priority consequently reads a cold plate as an unwanted one, and the units cold longest are the least likely to be replaced. None of the 82 cold units appeared on the current shortlist, which is drawn from the top two usage quintiles. Next: a demand estimate taken at the unit rather than from it.

Selected references

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We thank the reserve maintenance crews who granted out-of-hours access to the audited units.

Unit-level probe records only; no visitor observation was retained. Thresholds were pre-registered before the first audit.

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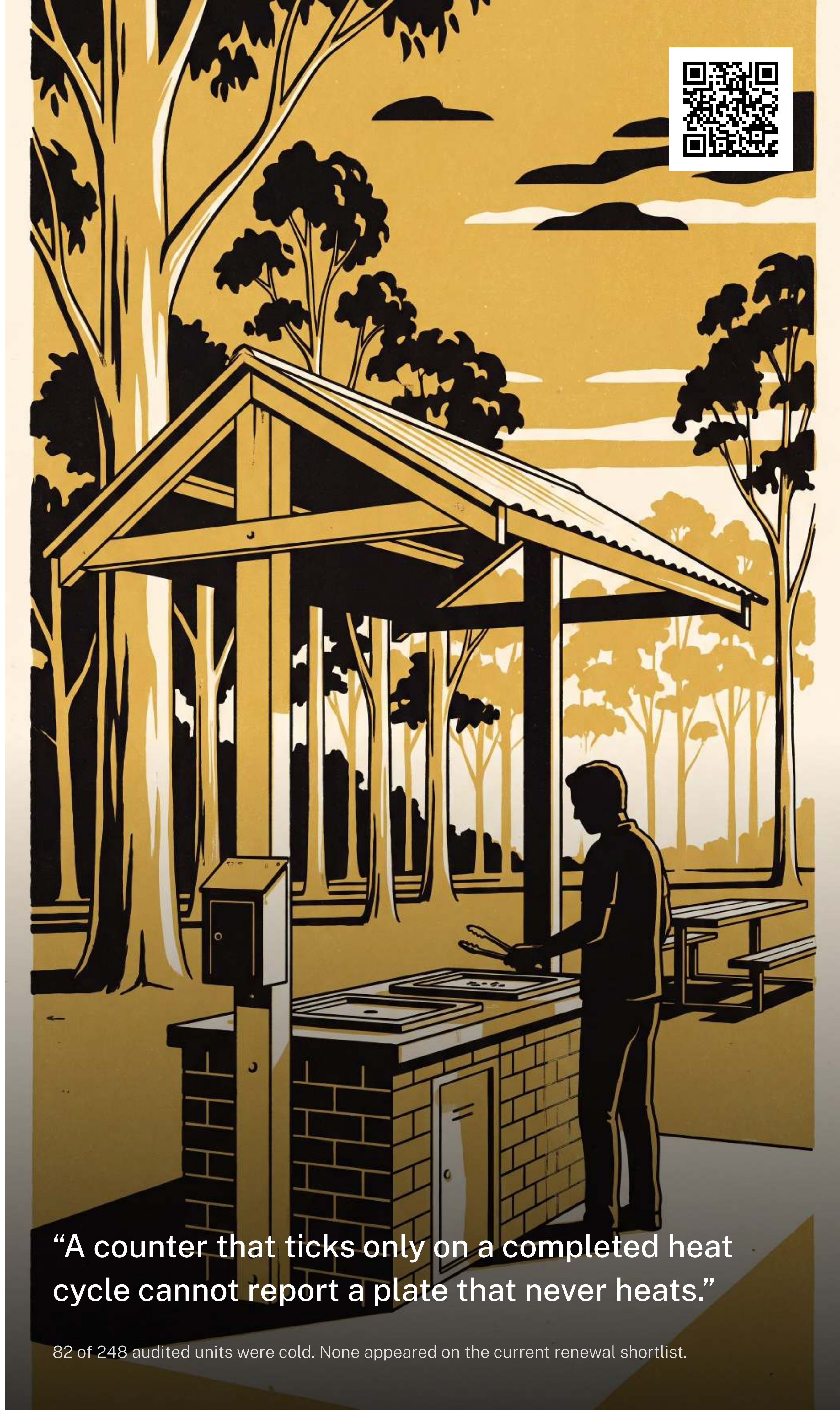

"A counter that ticks only on a completed heat cycle cannot report a plate that never heats."

82 of 248 audited units were cold. None appeared on the current renewal shortlist.

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